

CRISTOFF STAN

Senior Software Developer

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PROFESSIONAL SUMMARY

Senior Software Developer with 9+ years of experience building scalable backend systems and full-stack applications. Proven track record delivering Python and Django solutions in data-intensive domains including bioinformatics, analytics, and medical software. Adept in AWS, CI/CD pipelines, and high-performance workflows. Experienced with NGS data pipelines, RESTful APIs, PostgreSQL, and collaborative team development in remote international setups.

TECHNICAL SKILLS

- **Languages:** Python, TypeScript, JavaScript, Bash, SQL, HTML5, CSS3
- **Frameworks:** Django, FastAPI, React, Express.js, Node.js
- **Tools & Platforms:** AWS (EC2, S3), GitHub Actions, Docker, Git, Playwright, Pytest, PostgreSQL, MySQL, Linux, HPC environments
- **CI/CD & Testing:** GitHub Actions, Jenkins, CircleCI, Pytest, Playwright, Unit/Integration/E2E Testing
- **Data & Analysis:** Pandas, NumPy, SciPy, Scikit-learn, PostgreSQL, JSON Schema
- **Other:** RJSF, Figma, RESTful APIs, JIRA, Confluence

WORK EXPERIENCE

Sensidev

Senior Software Engineer

September 2021 – December 2024

Timișoara, Romania

- Designed and developed bioinformatics tools for oncology research using Python, Django, and Bash scripting to process high-throughput genomic data on distributed compute clusters.
- Engineered automation scripts for HPC environments, optimizing NGS pipelines across 10,000+ CPUs, reducing processing time by over 40%.
- Implemented robust API integrations and data management pipelines for clinical trial metadata and patient sequencing data using PostgreSQL and Django ORM.
- Built web-based frontend components using React and JSON Schema Forms (RJSF), collaborating with data scientists to improve visualization of genomic patterns.
- Deployed scalable services on AWS EC2 and S3, configuring environment isolation and high-performance batch operations with Bash and BDS.
- Developed test suites using Pytest and Playwright to ensure reliability and performance of backend and frontend pipelines across environments.
- Enhanced CI/CD workflows with GitHub Actions, automating testing, build, and deployment across dev, staging, and production environments.

- Collaborated cross-functionally in an agile team across Europe, contributing to product design, code reviews, sprint planning, and release strategy.

Syscom Digital

Software Engineer

October 2019 – August 2021

Bucharest, Romania

- Created Django-based RESTful APIs to integrate clinical devices with mobile interfaces for remote diagnostics and health reporting.
- Managed SQL databases for high-volume sensor data storage and retrieval, designing optimized queries for statistical and diagnostic reporting.
- Refactored legacy Python scripts into modular packages, reducing maintenance costs and improving test coverage using unit tests and continuous integration.
- Supported frontend team with TypeScript and React enhancements, focusing on user experience for healthcare analytics dashboards.
- Integrated cloud infrastructure with AWS S3 and Lambda functions for secure data processing and automation pipelines.
- Designed schema validation pipelines using JSON Schema, enhancing data integrity for medical form submissions and test reports.
- Wrote advanced Bash scripts to automate deployment, database backups, and periodic health checks for bioinformatics systems.
- Introduced monitoring and logging solutions with Prometheus and Grafana to track NGS pipeline performance and failures.

Soft Build

Software Developer

June 2016 – July 2019

Timișoara, Romania

- Developed scalable backend features for e-health platforms in Django, managing patient data ingestion and anomaly detection workflows.
- Maintained Python scripts to clean, filter, and analyze biomedical signals and imaging data for research projects in diagnostics.
- Wrote bash-driven ETL scripts to prepare large genomic datasets for pipeline processing and feature extraction.
- Implemented CI pipelines using Git and CircleCI for Python microservices that served clinical research teams.
- Designed database schemas and optimized SQL queries to handle millions of genomic records across PostgreSQL and MySQL backends.
- Contributed to frontend interfaces in Angular and React, focusing on forms, input validation, and visualization for molecular reports.
- Deployed backend services to cloud using Docker and AWS EC2, provisioning isolated environments for researchers and clinicians.
- Participated in user testing sessions and integrated feedback into iterative development cycles for bioinformatics tools.

Decima Digital

Software Developer

November 2015 – April 2016

Tallinn, Estonia

- Built lightweight Python web services to parse and analyze molecular data for early-phase clinical trials.
- Developed reusable modules in Django for genomic result visualization and versioned sample tracking.
- Created automated shell scripts for nightly runs of genomic validation jobs using CRON and system monitoring tools.
- Worked closely with UX designers to ensure frontend features in React were intuitive and performance-optimized for scientific use.
- Performed code audits and bug triaging across bioinformatics modules, improving code quality and readability by 60%.
- Integrated external APIs for data enrichment and annotation using RESTful interfaces.
- Managed deployments and DNS configurations for project-specific servers hosted on AWS.
- Delivered scientific support documentation alongside code for integration with research lab systems.

Borna Technology

Software Developer Intern

September 2014 – August 2015

Tallinn, Estonia

- Assisted in developing data pipelines for a medical device firmware backend, processing real-time vitals and generating alerts.
- Learned and contributed to Django-based microservices, focusing on task queues and asynchronous job handling.
- Conducted literature reviews and data cleaning efforts for integration of third-party biomarker datasets.
- Contributed to frontend prototypes using vanilla JS and HTML5 for remote lab data dashboards.
- Wrote internal documentation and conducted validation testing of bioinformatics tools used in diagnostics.
- Improved existing shell scripts used in preprocessing NGS data on local clusters, optimizing runtime.
- Participated in sprint meetings and code review sessions with senior engineers across distributed teams.
- Gained familiarity with Git, Linux, and PostgreSQL under real-world supervision.

EDUCATION

Tallinn University

Bachelor's Degree in Computer Science,
2010 – 2014

CERTIFICATIONS

- ❖ AWS Certified Solutions Architect – Associate (2023)
- ❖ Django for Professionals (Udemy, 2022)
- ❖ Bioinformatics Specialization – Coursera, 2021)
- ❖ SQL for Data Science – (Coursera, 2020)

PROJECTS

- ✓ **Precision Oncology Pipeline** – Built scalable pipelines to analyze next-generation sequencing data for biomarker discovery using Python, Pytest, PostgreSQL, and AWS HPC.
- ✓ **NGS Workflow Automation** – Developed shell-driven data orchestration for automated clinical data ingestion, normalization, and genomic signature extraction.
- ✓ **Genomic Report Generator** – Full-stack Django + React application for real-time visualization of molecular patient data, including exportable PDF summaries.
- ✓ **CI/CD Genomics Toolkit** – GitHub Actions-powered CI pipelines for data validation, deployment, and version management of internal genomics tools.